Lectures

STAT 109: Introductory Biostatistics

Readings

Below each module is a list of learning objectives, e.g. what you should be able to do when you finish that module.

Module 0 — Start here

Step	Material
Read 0	Module 0 Read 0 — welcome, statistical process, three worlds, five projects, Colab & R
Read	Syllabus · Course path · Resources
Quiz	Quiz 0 (Start Here) · Quiz 0B (Question Ideas)
Reference sheets	Methods map · R reference
Lab	Demo lab · Exercise lab
Canvas	Module 0 discussion (introduce yourself)

After Module 0 you should be able to:

- Navigate where readings, labs, and projects live vs what happens in Canvas.
- Open R in Google Colab, save a copy, and run notebook cells.
- Use variables, vectors, logical TRUE/FALSE, and a simple simulation loop.

Module 1 — Testing a Claim

Is this observation surprising?

Course path

Step	Material
Read 0	Module 1 Read 0 — project preview & sample tour
Read	Module 1 Read 1
Quiz A	Def & Notation
Read more	Module 1 Read 2
Quiz B	Concept Check
Demo lab	Module 1 demo · Conditional demo
Exercise lab	Module 1 exercise lab
R reference	R quick reference (Module 0–1)
Milestone	Project 1 milestone · Open in Colab
Synthesis	Module 1 synthesis
Project	Project 1

Read 1 — Probability through the binomial pmf

Essential · after Read 0

- Compute counts, proportions, and percentages; use x, n, $\hat{p}$, and $P(\text{event})$.
- Define sample space, outcome, and event; use the equally likely rule.
- Apply addition, complement, conditional, and multiplication rules; check independence.
- Use permutations and combinations; define binomial process and pmf.

Read 2 — Binomial test in practice

Essential · after Read 1 and Quiz A

- Translate a research question into H_0 and H_a; verify BRP conditions.
- Plot the null binomial distribution; mark observed x; compute and interpret a p-value.
- Carry out the full Project 1 workflow for one- and two-sided alternatives.

Module 2 — Estimating the Truth

Building high-quality estimates from samples.

Course path

Step	Material
Read 0	Module 2 Read 0 — project preview & sample tour
Read	Module 2 Read 1
Quiz A	Def & Notation
Read more	Module 2 Read 2
Quiz B	Concept Check
Demo lab	Module 2 demo — tools · worked
Exercise lab	Module 2 exercise lab
Milestone	Project 2 milestone · Open in Colab
Synthesis	Module 2 synthesis
Project	Project 2

Read 1 — One proportion: errors, exact binomial, and approximation

Essential · after Read 0

- Define **Type I** and **Type II** errors and relate **significance level** α to the risk of a false alarm.
- Contrast a **hypothesis test** (*is* p equal to a claimed p_0?) with a **confidence interval** (*what* values of p are plausible?).
- Use **`binom.test()`** for an exact binomial **p-value** and **95% CI**; interpret the CI in context.
- Use $\mu = np$ and $\sigma = \sqrt{np(1-p)}$; check $np \geq 10$ and $n(1-p) \geq 10$ before normal-based methods.
- State the **Empirical Rule**; compute **Wald** and **Plus Four** 95% CIs; use **`prop.test(x, n)`** when large-count conditions hold.
- Compute the **standardized test statistic** for a one-proportion test; contrast test SE (p_0) with CI SE ($\hat{p}$).

Read 2 — Sampling, one numerical variable, and FPC

Essential · after Read 1 and Quiz A

- Define **population**, **sample**, **parameter**, and **statistic**; describe **sampling distribution** and **confidence level**.
- Discuss **bias** vs **precision** (including role of sample size n); identify **SRS** vs **convenience** sampling.

- Identify a **sampling frame**, **carry out** SRS, and **classify** sampling designs (convenience, SRS, cluster, systematic, stratified, multi-stage).
- Compute **sample mean**, **sample SD** s, **standard error** $s/\sqrt{n}$, and a **95% CI for a mean** (by hand and with **`t.test()`**).
- Graph one numerical variable with **ggplot2** (**dot plot** and **histogram**); summarize with **`summarize()`**.
- State when to apply the **finite population correction (FPC)** and compute **95% CIs with FPC** for a proportion or mean.
- Use the **conditions checklist** and **Project 2 decision table**; complete a worked CI with the **interpretation recipe**.

Optional for Project 2: one-sample `t.test()` for hypothesis testing (CI is the project focus). *Optional enrichment:* z-scores, methods-map toolbox.

Module 3 — Discovering Patterns

Describing a dataset with graphs and summaries.

Course path

Step	Material
Read 0	Module 3 Read 0 — descriptive toolbox & project preview
Read	Module 3 Read 1 — one categorical, one numerical, numerical × categorical
Quiz A	Def & Notation
Read more	Module 3 Read 2 — two categorical, two numerical
Quiz B	Concept Check
Demo lab	Module 3 demo hub · Tools · Worked
Exercise lab	Module 3 exercise lab
Milestone	Project 3 milestone · Open in Colab
Synthesis	Module 3 synthesis
Project	Project 3

Read 1 — One categorical, one numerical, numerical by group

Essential · after Read 0

Section 1 — One categorical: counts, proportions ($\hat{p} = x/n$), bar charts and pie charts; same x, n notation as Modules 1–2.

Section 2 — One numerical: histograms and dot plots; **center**, **spread**, **shape**, **unusual observations**; mean, median, SD, MAD, range, quartiles, IQR, five-number summary, boxplot.

Section 3 — Numerical × categorical: compare groups with **`group_by()`** + **`summarize()`**, side-by-side boxplots/histograms; Welch t-test **optional**.

Read 2 — Two categorical and two numerical

Essential · after Read 1 and Quiz A

Section 4 — Two categorical: two-way tables; joint, row, and column proportions; stacked bars (counts and `position = "fill"`).

Section 5 — Two numerical: scatterplots; **form**, **direction**, **strength**; **Pearson correlation** r when form is roughly linear.

Module 4 — Designing Fair Comparisons

Comparing two groups responsibly.

Course path

Step	Material
Read 0	Module 4 Read 0 — project preview & fair-comparison vocabulary
Read	Module 4 Read 1 — study design · two-group tests/CIs · more than two groups
Quiz A	Def & Notation
Demo lab 1	Simulate & compare groups
Demo lab 2	Tests & confidence intervals
Exercise lab	Module 4 exercise lab
Milestone	Project 4 milestone
Synthesis	Module 4 synthesis
Project	Project 4

Read 1 — Study design, two-group inference, and comparing more than two groups

Essential · after Read 0 · three sections

Section 1 — Study design (including confounders)

- Compare **randomized controlled trials** and **observational** designs when the goal is a causal claim about a treatment or exposure.
- List the **three explanations** for an observed difference: **causation, chance, confounding**.
- Define a **confounder** in words and in a **DAG**, and use **subclassification** to compare groups within levels of a measured confounder.
- Distinguish **random assignment** from **random sampling**.

Section 2 — Two-group tests and confidence intervals

- Pick `t.test()` for two independent numerical groups or `prop.test()` / `chisq.test()` for binary comparisons, and name key conditions.
- Interpret **Welch two-sample *t*-test** output (hypotheses, p-value, confidence interval) in context.
- Recognize **paired/matched** data and use the **paired *t*-test** or **McNemar's test**.

Section 3 — Comparing more than two groups

- Use **one-way ANOVA** (`lm()` then `anova()`) for a numerical outcome across 3+ groups, and the **chi-square test of independence** (`chisq.test()`) for a categorical/binary outcome across groups.
- Explain why **Project 4 keeps to two groups**.

Module 5 — Looking Ahead

Using data to make predictions.

Course path

Step	Material
Read 0	Module 5 Read 0 — project preview & four-day schedule
Read	Module 5 Read 1

Step	Material
Quiz A	Def & Notation
Read more	Module 5 Read 2
Quiz B	Concept Check
Demo lab	Module 5 demo
Exercise lab	Module 5 exercise lab
Synthesis	Module 5 synthesis
Milestone	
Project	Project 5 · **Portfolio package due Thu Jul 2**

Read 1 — Correlation and the least-squares line

Essential · after Read 0

- Read a scatterplot using **form** (linear or curved), **direction** (positive or negative), and **strength** (strong, moderate, or weak).
- Compute and interpret **Pearson correlation** r as a measure of strength and direction of a linear association.
- Fit a **simple linear regression** line $\hat{y} = b_0 + b_1 x$ with `lm()` and identify slope b_1 and intercept b_0.
- Interpret the **slope** in context: predicted change in y for a one-unit increase in x.
- Use the fitted equation to make a **point prediction** for a new x value.
- Define a **residual** as observed minus predicted $(y - \hat{y})$ and explain what the least-squares criterion minimizes.
- Interpret **MSE** and R^2 on a test set as summaries of predictive performance on new data.

Read 2 — Inference in simple linear regression

Essential · after Read 1 and Quiz A

- State the residual-normality assumption that supports interval inference in linear regression.
- Interpret a **95% confidence interval for the slope** β_1 in context.
- Interpret a **confidence interval for the mean response** at a specific x_0 value.
- Interpret a **prediction interval** for an individual future response at x_0.
- Explain why **prediction intervals** are wider than confidence intervals for the mean response.
- Use `confint()` and `predict(..., interval = "prediction")` in R to extract these intervals.